- **Named Ranges**: You discovered how naming a range of cells simplifies referencing them in formulas across your spreadsheet. For instance, after naming a price column as house_prices, you can easily use it in calculations like =house_prices/area to find the price per unit area.

=house_prices/area

- **Subtotals**: You explored using the Subtotal feature for quick calculations, such as finding the average or sum of a column based on specific criteria. This helps in analyzing data sets without manually calculating each value.
- **Data Validation**: You learned to ensure data integrity by restricting the type of data that can be entered into a spreadsheet. For example, setting a column to only accept whole numbers prevents incorrect data entries.
- **Custom Formats**: You found out how to create custom formats for data presentation, making your data easier to read and understand. This includes formatting dates, numbers, and even setting conditional formatting to highlight specific data points.
- **Conditional Formatting**: You utilized conditional formatting to visually distinguish and highlight significant data points, such as the top 100-priced houses in a dataset, using color scales, data bars, or icons.

Visualizing disaggregated data with PivotCharts

You learned about the power of PivotTables and PivotCharts in Excel for visualizing disaggregated data, which refers to detailed, raw data that hasn't been summarized yet. Here's a quick recap of the key points:

- **Aggregated vs. Disaggregated Data**: You explored the difference between aggregated data (summarized information like total sales per region) and disaggregated data (detailed, transaction-level data).
- **Introduction to PivotTables and PivotCharts**: You discovered how PivotTables can organize and summarize your data, and how PivotCharts provide dynamic visualizations that update with changes to the PivotTable.

Creating a PivotChart: You learned to create a PivotChart by selecting a dataset and using the PivotChart tool. For example, to visualize the sum of songs per year from a

dataset, you would:
Insert > PivotChart > Select data range > Choose PivotChart location

- **Customizing PivotCharts**: You customized your PivotCharts, changing chart types and adding data fields to enhance your data analysis.
- **Building a Mini Dashboard**: You combined multiple PivotCharts into a dynamic dashboard, incorporating slicers and timelines for interactive data exploration.
- **Printing Dashboards**: Finally, you learned about Excel's printing options to share your dashboards, including setting print areas, adjusting page layouts, and ensuring your visualizations fit on one page.
- **Scenario Analysis**: You discovered that scenario analysis evaluates a dependent variable's performance based on specific input variables. It's useful for answering "What would happen if...?" questions, such as calculating potential taxes based on different incomes, deductions, and tax rates.
- **Independent vs. Dependent Variables**: You learned the difference between these two types of variables. Independent variables are the inputs of your analysis, while dependent variables are the outputs that depend on the inputs.
- **Sensitivity Analysis**: This type of analysis was explained as evaluating how a dependent variable responds to a range of inputs, unlike scenario analysis, which focuses on specific scenarios. Sensitivity analysis often uses tables to display how outcomes vary with different input values.
- **Growth Rates Calculation**: You learned how to calculate growth rates by finding the percent change between two values. For example, to calculate the growth rate from 50 million to 70 million, you subtract 50 million from 70 million to get 20 million, then divide by 50 million to get a 40% growth rate.
- **Excel Tools for What-If Analysis**: Several Excel tools were introduced for performing What-If Analysis, including the Goal Seek and Scenario Manager tools. You saw how these tools can help in achieving specific goals or analyzing multiple scenarios by adjusting input values.
- **Practical Application**: Through exercises, you applied what you learned by creating scenario analyses, using Goal Seek, and setting up sensitivity analyses with Data Tables in Excel. For instance, you used the formula =[@[Upsell Opportunity]]*[@[Conversion Rates]] to calculate projected upsell amounts under different scenarios.
- **Seasonality**: You discovered how seasonality affects data trends, with examples like retail sales peaking in December.
- **Bias in Data**: You explored different types of biases such as sampling bias, confirmation bias, and anchoring bias, and learned their impacts on forecasting accuracy.

- **Confidence Intervals**: You understood that confidence intervals provide a range within which a future outcome is likely to occur, with higher confidence levels leading to wider intervals.
- **Moving Averages**: You learned how moving averages smooth out data to identify trends, and the difference between simple and weighted moving averages. For example, a weighted average calculation might look like this: =B2*15%+B3*35%+B4*50%.
- **Forecasting in Excel**: You practiced using Excel's forecasting tools, including setting up moving averages, adding trendlines to charts, and utilizing the Forecast Sheet feature for sophisticated predictions that consider seasonality and confidence levels.
- **Creating Calculated Fields**: You started by calculating the churn rate, using a formula to measure the percentage of customers leaving. For example, Churn Rate % = 'Churned Customers' / 'Total Customers'.
- **Analyzing Demographics**: You learned to categorize customers into demographics using Excel formulas, such as =IF([@[Under 30]]="Yes", "Under 30", IF([@Senior]="Yes", "Senior", "Other")), and then analyzed the churn rate within these groups.
- **Segmenting by Age and Plans**: You segmented customers by age and analyzed the impact of unlimited data plans on churn, discovering that senior citizens churn more often and that data plan types influence customer loyalty.
- **International Calls and Contract Types**: You investigated how international calling plans and different contract types affect customer loyalty, using PivotTables to organize and analyze the data.
- **Advising on Retention Strategies**: Based on your findings, especially the high churn rate among senior citizens, you considered strategies to enhance customer retention and improve Databel's competitiveness.
- **Importance of Clean Data**: Understanding that clean data is vital for ensuring accuracy, efficiency, effectiveness, consistency, and integrity in data analysis.
- **Handling Errors**: You discovered how to identify and correct errors in datasets, such as incorrect dates or doses, using Power Query's error handling options like deleting or replacing errors.
- **Removing Duplicates**: The lesson covered the importance of removing duplicate data to maintain the dataset's integrity and how to identify and eliminate these duplicates in Power Query.
- **Dealing with Missing Data**: You learned strategies for identifying and correcting missing data, such as replacing null values with specified defaults to ensure completeness of the dataset.
- **Outlier Detection**: Techniques for identifying and correcting outliers in survey responses were discussed, ensuring that data falls within expected ranges.

For example, to replace errors in the `birth_date` column, you would:

Right-click on the column > Replace Errors > Enter correct value

Power Query/PowerPivot

- **Define Clear Objectives**: A dashboard should answer specific questions about the data.
- **Know Your Audience and Scope**: Tailor the dashboard to the needs of its users.
- **Keep It Simple and Focused**: Avoid clutter to maintain clarity.
- **Balance Aesthetics with Functionality**: Ensure the dashboard is both visually appealing and easy to use.

- Click on KPIs, then New KPI.

- Define the base field and target value.

- Set status thresholds (e.g., Green for 100%, Yellow for 90-99%, Red below 90%).

- Select an icon style and click OK.